

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

Nov 2012

EE309 Electrical Power Utilization & Traction

Date: 08.11.2012, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 What is Crest speed, Average speed and Schedule speed? Also explain the factors affecting schedule speed. [07]

Q - 2 (a) Draw Trapezoidal speed-time curve and derive the equation for this curve. [07]

(b) An electric train has an average speed of 42 kmph on a level track between stops 1,400 m apart. It is accelerated at 1.7 kmphs and braked at 3.3 kmphs. Draw the speed time curve for the run and find out Acceleration, free running and Braking periods. [07]

OR

Q - 2 (a) What is Load Equalization? Derive formula $T_m = T_L - (T_L - T_0) e^{-\frac{t}{KT}}$ from first principle. [07]

(b) What do you mean by Traction and Electric Traction System? Give classification of traction systems. And what are the fulfill requirements of an ideal traction system. [07]

Q - 3 (a) Explain any three speed control methods for 3-phase induction motor with neat diagram. [06]

(b) Explain running characteristics of DC shunt motor with necessary diagrams. [08]

OR

Q - 3 (a) Write in brief about Direct core type Induction Furnace. [05]

(b) Write a short note on Gas-filled lamp. [05]

(c) Write a short note on Halogen lamp. [04]

SECTION – II

Q - 4 What is electrolytic process? Explain its basic principle with neat diagram. [07]

Q - 5 (a) Define the following terms. [04]

(i) Lumen (ii) Lux (iii) Candela (iv) Illumination

(b) Compare between Resistance Welding and Arc Welding [04]

(c) Explain different modes of Transfer of heat. [06]

OR

Q - 5 (a) What are the various factors to be considered in selection of an Electric Motor for a particular service? [06]

(b) A 220 V dc shunt motor takes 22 A at rated voltage and runs at 1000 rpm. Its field resistance is $100\ \Omega$ and armature circuit resistance is $0.1\ \Omega$. Compute the value of additional resistance required in the armature circuit to reduce the speed to 800 rpm when (i) the load torque is proportional to speed and (ii) the load torque varies as the square of the speed. [08]

Q - 6 (a) Write down advantages of electric heating over other systems of heating. [08]

(b) Write down Applications and advantages of Eddy current heating. [06]

OR

Q - 6 (a) An electric train weighing 200 tonne has eight motors geared to driving wheels, each wheel is 90 cm diameter. Determine the torque developed by each motor to accelerate the train to a speed of 48 kmph in 30 seconds up a gradient of 1 in 200. The tractive resistance is of 50 newtons per tone, the effect of rotational inertia is 10 % of the train weight, the gear ratio is 4 to 1 and gearing efficiency is 80 %. [06]

(b) Compare between Tungsten filament lamps and Fluorescent tubes. [08]
